

Problems on Laplace Transformation

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Notation: $\mathcal{LT}\{f(t)\} = F(s)$ = Laplace Transform of the function $f(t)$.

Question 1. (a) Let n be a non negative integer and a be a real constant. Show that $f(t) = t^n \cdot \sin(at)$ is exponentially bounded.

(b) Using Laplace Transforms, show that e^{t^2} is not exponentially bounded.

(c) Let $f(t)$ be a continuous function on $[0, \infty)$ such that $\lim_{t \rightarrow \infty} e^{-st} \cdot f(t)$ is finite, for some $s \in \mathbb{R}$. Show that $\mathcal{LT}\{f(t)\}$ exists.

Question 2. (a) Can the function $F(s) = \left(1 + \frac{a}{s}\right)^s$ be the Laplace Transform of a function $f(t)$? If yes, evaluate $f(0)$. If no, why?

(b) Let $f(t) = \cosh t$. Given that $\mathcal{LT}\{f(t)\} = \frac{s}{s^2 - 1}$ for $s > 1$, evaluate $\mathcal{LT}\{g(t)\}$ where $g(t) = tf(t)$.

(c) How can we modify $f(t)$ to obtain a function $h(t)$ such that $\mathcal{LT}\{h(t)\} = s \cdot \mathcal{LT}\{f(t)\}$?

Question 3. Given $J_0(t) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2} \left(\frac{t}{2}\right)^{2m}$, then prove that $\mathcal{LT}\{J_0(t)\} = \frac{1}{\sqrt{1+s^2}}$

Question 4. Obtain the Laplace transform of the function $\int_2^t f(\tau) d\tau$ in terms of $\mathcal{LT}\{f(t)\} = F(s)$, given that $\int_0^2 f(\tau) d\tau = 3$.

Question 5. If $\mathcal{LT}\{f(t)\} = F(s)$, then prove that $\mathcal{LT}\left\{\int_0^t \int_0^t f(t) dt\right\} = \frac{1}{s^2} \mathcal{LT}\{f(t)\}$.

Question 6. (a) Let $f(t)$ be a piecewise continuous periodic function on $[0, \infty)$ with fundamental period T . Then show that

$$\mathcal{LT}\{f(t)\} = F(s) = \frac{1}{1 - e^{-Ts}} \int_0^T e^{-st} \cdot f(t) dt$$

(b) Find the Laplace Transform of the following function:

$$f(t) = e^{-t}, \quad 0 \leq t \leq 2 \quad \text{and} \quad f(t+2) = f(t)$$

(c) Let

$$g(t) = \begin{cases} e^{20t}, & 0 \leq t < \pi \\ \sin t, & \pi \leq t < 2\pi \\ g(t - 2\pi), & t \geq 2\pi \end{cases}$$

Evaluate the integral $\int_0^\infty e^{-2025t} g(t) dt$

Question 7. Write the following function using unit step functions and find its Laplace transform.

$$f(t) = \begin{cases} \sin(\pi t), & 2 \leq t < 4, \\ e^{-\pi t}, & 4 \leq t < 6, \\ t^2, & 8 \leq t < 10, \\ 0, & \text{otherwise.} \end{cases}$$

Question 8. (a) Evaluate the integral $\int_0^\infty \int_0^t e^{-36t} \cdot \frac{\cos \tau}{e^{-\tau}} d\tau dt$.

(b) Find the Laplace Transform of

$$f(t) = \begin{cases} \int_{2\pi}^{3\pi} h(p) dp + \int_{4\pi}^{5\pi} h(p) dp + \cdots + \int_{2r\pi}^t h(p) dp, & r = 1, 2, \dots, 2r\pi \leq t < (2r+1)\pi \\ \int_{2\pi}^{3\pi} h(p) dp + \int_{4\pi}^{5\pi} h(p) dp + \cdots + \int_{2r\pi}^{(2r+1)\pi} h(p) dp, & r = 1, 2, \dots, (2r+1)\pi \leq t < (2r+2)\pi \\ 0, & t \leq 2\pi \end{cases}$$

where, $h(p) = \sin(t-p) \cdot p \cos p$

Question 9. (a) State the hypothesis required for first and second limit theorems.

(b) Find a function $f(t)$ such that $\lim_{s \rightarrow 0} sF(s)$ exists, but $\lim_{t \rightarrow \infty} f(t)$ does not exist. Explain why the second limit theorem cannot be applied in this case.

Question 10. (a) Let

$$F(s) = \frac{k_1 s + k_2}{s(s+2)(s+a)},$$

where $a > 0$. Determine the values of k_1 and k_2 such that $\lim_{t \rightarrow \infty} f(t)$ exist and is finite. Also, compute this value in terms of k_1, k_2 and a .

(b) Find $\lim_{t \rightarrow 0^+} f(t)$ and $\lim_{t \rightarrow \infty} f(t)$ when

$$F(s) = \frac{s^2 + 4s + 5}{s(s+1)(s+2)},$$

provided these limits exist and Laplace transform of $f'(t)$ exist.

Question 11. Consider the initial value problem

$$y'' + y' + y = f(t), \quad y(0) = y'(0) = 0 \quad t > 0$$

where

$$f(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ -1, & 1 < t < 2 \end{cases} \quad f(t+2) = f(t)$$

Determine $Y(s)$.